

MIRUNALINI A

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PROFILE

Aspiring Electronics & Embedded Systems Engineer pursuing B.Tech in Electronics and Computer Engineering at VIT Chennai, with hands-on experience across embedded firmware (ESP32), FPGA/digital design (Verilog HDL), edge AI, computer vision, IoT systems, and control systems (MATLAB/Simulink). Holder of one published and one filed patent in ML-driven hardware systems.

EDUCATION

Vellore Institute of Technology, Chennai B.Tech – Electronics and Computer Engineering CGPA: 8.59	2023 – 2027
Everwin Vidhyashram (CBSE) Class 12 – Computer Science Stream: 91% Class 10: 85%	2019 – 2023

WORK EXPERIENCE

Ikomet Technologies <i>AI & Software Engineering Intern</i> - Contributed to the development, testing, and evaluation of a RAG-based SAT Question Generation System using Generator–Validator AI agents. - Performed functional testing and QA validation of AI-generated questions, identifying content, logic, and workflow issues to improve system accuracy and reliability. - Reviewed AI-driven learning platform content and UI/UX, and researched UK education curricula and localization requirements to support product expansion.	Jun 2026 – Jul 2026
Qmax Test Equipments Pvt Ltd <i>Embedded & IC Testing Intern</i> - Worked with Automated Test Equipment (ATE) for IC model programming and AC/DC parametric testing using Python scripts. - Interpreted IC datasheets and performed functional verification using IEEE VHDL and Python, automating test routines to improve efficiency. - Applied graphical "learn and compare" techniques for IC validation, contributing to diagnostic analysis workflows.	May 2025 – Jun 2025
Heavy Vehicles Factory, Avadi, Chennai <i>Implant Trainee</i> Gained practical exposure to system validation, parts testing, and assembly line workflows in a large-scale manufacturing environment.	Dec 2024 – Jan 2025

PROJECTS

AgroBot — ML-Powered Precision Agriculture Robot — <i>Raspberry Pi 5, Python, ML, Computer Vision, Mobile App</i> - Developed a CV-based plant disease classification pipeline on Raspberry Pi 5, triggering precision-targeted pesticide spraying to reduce chemical wastage in the field. - Built a companion mobile app for live robot monitoring, control, and field analytics; selected for Smart India Hackathon 2025 (SIH'25). Patent published for the detection and spraying mechanism.
Maternal-Fetal Health Monitoring and Risk Prediction System — <i>ESP32-S3, Edge AI, Wireless Sensor Fusion, IoT</i> - Designed an edge-AI health monitoring system on ESP32-S3 for real-time prediction of fetal distress and preterm labour, integrating wireless multi-sensor fusion for continuous tracking. - Performed inference at the edge to minimize reliance on cloud round-trips, enabling low-latency, safety-critical monitoring. Patent filed for the prediction system.
12×12 Multiplier using Hybrid Adder — <i>Verilog HDL, FPGA, Digital Design</i> - Designed a 12×12 multiplier combining Carry Look-Ahead and Carry Select adder architectures to optimize the speed–power trade-off in digital multiplication. - Simulated and verified the design in Verilog HDL and implemented it for FPGA deployment to validate timing and functional correctness.
AI-Enabled Wastewater Treatment & Predictive Control System — <i>Transformer Neural Network, ONNX, MATLAB/Simulink, MPC</i> - Trained a Transformer neural network to forecast key wastewater treatment process parameters from historical data, enabling multistep prediction of plant behavior. - Exported the model to ONNX and integrated it into MATLAB/Simulink via the Deep Learning Toolbox Predict block, driving a Model Predictive Controller for real-time process optimization.
Remote Health Monitoring System — <i>ESP32, Embedded C, WiFi, Firebase, React Native</i> - Developed ESP32 firmware for continuous multi-sensor (heart rate, SpO2, temperature) acquisition with interrupt-driven, real-time processing. - Implemented WiFi-based Firebase cloud communication for real-time data sync, paired with a React Native app and web dashboard for remote monitoring and alerts.

PATENTS

Closed-Loop Liquid Dispensing System with Image-Based Spatial Targeting and Deposition Verification — Patent Application Published (App. No. 202641065168, Pub. No. 23/2026); ML-based plant disease detection and precision-targeted spraying mechanism (AgroBot).
Maternal-Fetal Health Monitoring and Risk Prediction System — Patent Application Filed; edge-AI based real-time prediction of fetal distress and preterm labour using wireless sensor fusion.

TECHNICAL SKILLS

Embedded & IoT: ESP32, Arduino, Embedded C, VHDL, Verilog, PCB Design, LTspice
Data Science & ML: Python, NumPy, pandas, scikit-learn, TensorFlow Lite, Machine Learning, Signal Processing, MATLAB/Simulink
Programming Languages: Python, Java, JavaScript
Web & Mobile: HTML, CSS, JavaScript, Flask, React Native, Firebase
Tools & Fundamentals: Git, Jupyter Notebook, OOPS, DSA, Operating Systems, Database Concepts

CERTIFICATIONS & EXTRA-CURRICULAR

Spoken Tutorial, IIT Bombay – ns-3 Network Simulator Training, VIT Chennai (Feb 2025)
EduPyramids, SINE IIT Bombay – Java Training, VIT Chennai (Nov 2025)
EduPyramids, SINE IIT Bombay – Advanced C++ Training, VIT Chennai (Nov 2025)
WEBDEX Hackathon – Website on Sustainable Development, ACM SC VITC (Mar 2024)
Hands-on Workshop on Vega Processors and Ecosystem, VIT Chennai (Feb – Apr 2025)
Udemy – Learn the Art and Science of PCB Design with Eagle (Jul 2024)
Marketing Member, CodeChef Club – VIT Chennai

LANGUAGES KNOWN

English (Fluent) Tamil (Fluent) French (Basics)
